

✓ 1. Gumbel distribution

Using parameters (A, α) (location A , scale $\alpha > 0$):

$$f(x) = \frac{1}{\alpha} \exp\left(-\frac{x-A}{\alpha}\right) \exp\left(-\exp\left(-\frac{x-A}{\alpha}\right)\right)$$

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✓ 2. Log-likelihood under LDM

Let:

$$y_t = x_t - \theta t$$

Then:

$$\log \mathcal{L}^{X_t, LDM}(\theta; A, \alpha) = \sum_{t=1}^T \log f(y_t)$$

✓ Final expression

$$\ell(\theta, A, \alpha) = -T \log \alpha - \sum_{t=1}^T \frac{y_t - A}{\alpha} - \sum_{t=1}^T \exp\left(-\frac{y_t - A}{\alpha}\right)$$

✓ 3. First-order derivatives

Define:

$$u_t = \frac{y_t - A}{\alpha}, \quad e_t = \exp(-u_t)$$

◆ w.r.t. θ

$$\frac{\partial \ell}{\partial \theta} = \frac{1}{\alpha} \sum_{t=1}^T t - \frac{1}{\alpha} \sum_{t=1}^T t e_t$$

◆ w.r.t. A

$$\frac{\partial \ell}{\partial A} = \frac{T}{\alpha} - \frac{1}{\alpha} \sum_{t=1}^T e_t$$

◆ w.r.t. α

$$\frac{\partial \ell}{\partial \alpha} = -\frac{T}{\alpha} + \frac{1}{\alpha^2} \sum_{t=1}^T (y_t - A) - \frac{1}{\alpha^2} \sum_{t=1}^T (y_t - A) e_t$$

✓ 4. Second-order derivatives

◆ $\theta\theta$

$$\frac{\partial^2 \ell}{\partial \theta^2} = -\frac{1}{\alpha^2} \sum_{t=1}^T t^2 e_t$$

◆ AA

$$\frac{\partial^2 \ell}{\partial A^2} = -\frac{1}{\alpha^2} \sum_{t=1}^T e_t$$

◆ $\alpha\alpha$

$$\begin{aligned} \frac{\partial^2 \ell}{\partial \alpha^2} = & \frac{T}{\alpha^2} - \frac{2}{\alpha^3} \sum_t (y_t - A) \\ & + \frac{2}{\alpha^3} \sum_t (y_t - A) e_t - \frac{1}{\alpha^4} \sum_t (y_t - A)^2 e_t \end{aligned}$$

◆ Cross derivatives

θ - A

$$\frac{\partial^2 \ell}{\partial \theta \partial A} = -\frac{1}{\alpha^2} \sum_{t=1}^T t e_t$$

θ - α

$$\frac{\partial^2 \ell}{\partial \theta \partial \alpha} = -\frac{1}{\alpha^2} \sum_t t + \frac{1}{\alpha^2} \sum_t t e_t - \frac{1}{\alpha^3} \sum_t t (y_t - A) e_t$$

A - α

$$\frac{\partial^2 \ell}{\partial A \partial \alpha} = -\frac{T}{\alpha^2} + \frac{1}{\alpha^2} \sum_t e_t - \frac{1}{\alpha^3} \sum_t (y_t - A) e_t$$